APICURON: Database for Crediting & Recognition of biocuration activities

APICURON
The platform to Credit & Acknowledge scientific contributions.

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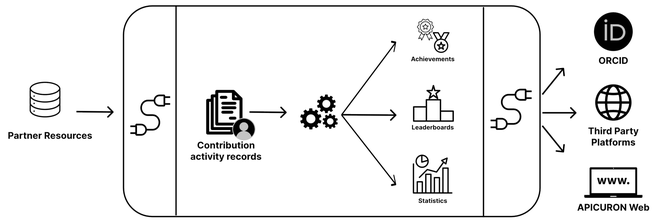
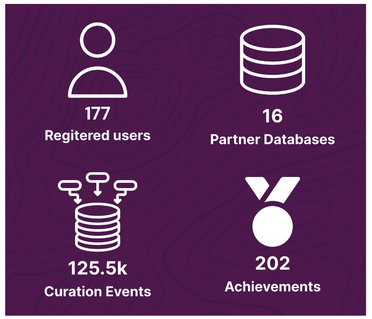
APICURON EXPLAINED



APICURON is a software-as-a-service based solution that aims to simplify effort recognition and help the scientific community recognize each contribution. It achieves so by serving the following purposes

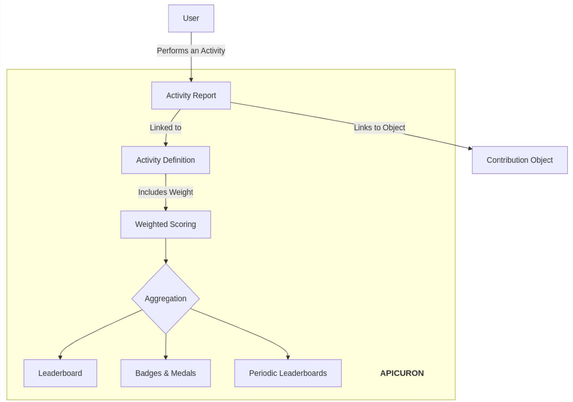
- Credit and Acknowledge scientific contributions effort
- A standardized **effort recognition system**
- **Serve as a Portfolio platform** for Scientists beyond publications
- Simple **feedback-loop** for platforms to evaluate **contributor engagement**



APICURON has shown **staggering success** in the scientific community integrating multiple scientific resources (from 3 to 16 resources) in 2 years, including scientific databases from the **EBI** (e.g **IntAct**, **Reactome**...etc).

With 177 registered users, 16 partner databases, and over 125,500 curation events tracked, APICURON is actively recognizing contributions across the scientific community. To date, it has awarded 202 achievements, highlighting the impact and dedication of biocurators worldwide.

FROM CONTRIBUTIONS TO RECOGNITION



A Flexible Credit System for Tracking Contributions

APICURON offers a customizable framework to recognize and reward contributions across domains like biocuration, analysis, and open-source projects.

Activities (Contributions)

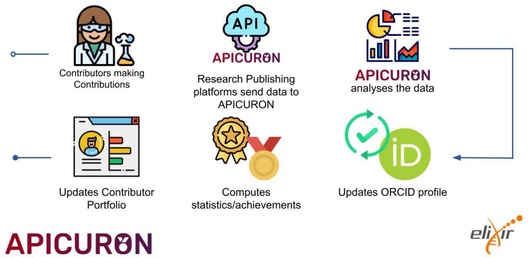
Defines credited actions (e.g., "contributing to documentation"), each with a weighted value to reflect its impact. Fully adaptable to different domains.

Activity Reports

Track user activities by linking the user, the action, and the contribution object (e.g., a URL). Enables detailed monitoring of individual contributions.

Reward Mechanisms & Achievements

APICURON supports flexible rewards that encourage participation and recognition.



Badges

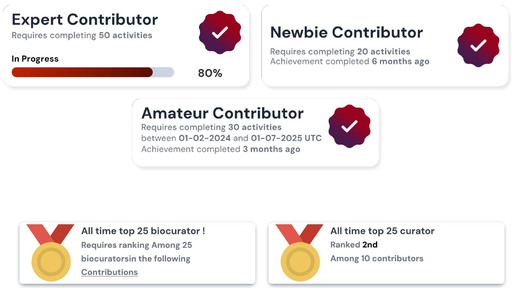
User-centered rewards that track progress and achievement dates. Can be updated continuously and set with time bounds to boost motivation.

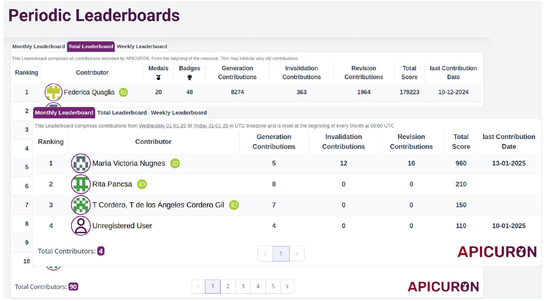
Medals

Ranking-based rewards appealing to competition. Users compete for top spots based on defined criteria. Time bounds enhance engagement and reduce discouragement.

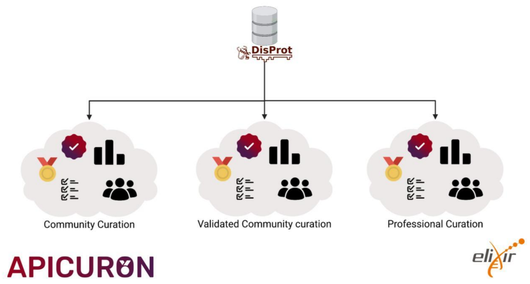
Leaderboards

Highlight recent contributions with time-based rankings. Weekly and monthly leaderboards reflect update cycles and encourage ongoing activity.





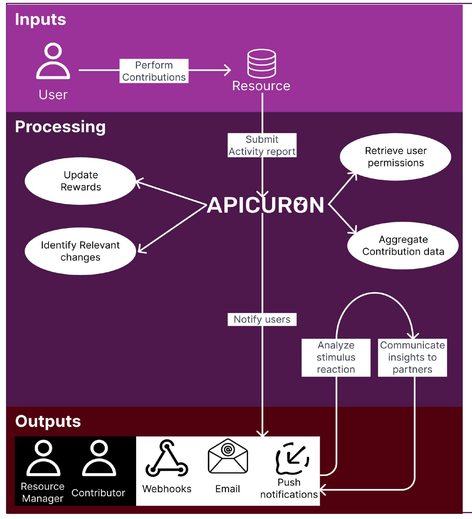
USE CASE: Managing contributor types via contribution groups



Contribution Groups in APICURON

*Use case: Separating **validated contributions** from **unvalidated contributions**.*

Contribution groups allow **registered contributors** to be organized into distinct **groups**. Each group can have its own set of **activities** and **reward mechanisms**, enabling resources to **differentiate** between contributor types, such as **community members** and **professionals**. For example, **DisProt** uses contribution groups to separate **validated contributions** from **independent efforts**. This setup promotes **fair competition** by tailoring leaderboards and achievements to each group.



FUTURE WORK: TOWARD AN ADVANCED NOTIFICATION AND REWARD SYSTEM

Advanced notifications System

APICURON offers valuable features to the **scientific community** but currently lacks **reactivity**. A key future goal is implementing an **advanced notification system** to help resource maintainers evaluate **credit and recognition strategies**. By triggering notifications and analyzing responses, we aim to **optimize fairness and effectiveness**.

However, this raises **ethical** and **data privacy** concerns. Early feedback from users and partners emphasized the importance of **transparent decision-making**. APICURON already follows a **privacy policy**, which will evolve alongside this development.

We also plan to introduce **additional reward mechanisms** that offer **greater specificity**, aligning with the diverse needs of our **early adopters**.

REFERENCES & ACKNOWLEDGMENTS

Hatos A, Quaglia F, Piovesan D, Tosatto SCE. APICURON: a database to credit and acknowledge the work of biocurators. Database (Oxford). 2021;2021:baab019. doi:10.1093/database/baab019

Ontiveros-Palacios N, Cooke E, Nawrocki EP, et al. Rfam 15: RNA families database in 2025. Nucleic Acids Res. 2025;53(D1):D258-D267. doi:10.1093/nar/gkae1023

Attafi OA, Clementel D, Kyritsis K, et al. DOME Registry: implementing community-wide recommendations for reporting supervised machine learning in biology. Gigascience. 2024;13:giae094. doi:10.1093/gigascience/giae094

Aspromonte MC, Nugnes MV, Quaglia F, et al. DisProt in 2024: improving function annotation of intrinsically disordered proteins. Nucleic Acids Res. 2024;52(D1):D434-D441. doi:10.1093/nar/gkad928

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